

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India 09/2021 – 01/2025 (3.5 Years)

- Developed and iterated on **rapid prototypes** for new robotics concepts, integrating **sensing, compute, connectivity, and motion** subsystems to de-risk system feasibility across diverse hardware and software components.
- Designed and executed **characterization, benchmarking, and test automation** for robotics platforms, generating performance and robustness data that informed key **architecture choices** and technology readiness levels.
- Built end-to-end **platform architectures** and reference implementations, creating reusable building blocks for future robotics solutions by integrating disparate hardware and software elements.
- Authored comprehensive **design docs, integration guides, and enablement materials** to transfer system learnings to product teams, clearly capturing tradeoffs, pitfalls, and recommended **system patterns** for complex robotics applications.
- Automated data collection and analysis for prototype testing using **Python scripting**, reducing manual effort by an estimated **40%** and accelerating iterative development cycles.
- Managed production incidents for enterprise **Java Spring Boot** applications, ensuring adherence to SLAs and coordinating resolution efforts for P1/P2 issues across distributed systems.

SKILLS

Robotics Software & Controls – ROS/ROS2, Scripting for automation and data analysis, System integration, Connectivity, Telemetry, Edge compute integration

Hardware & Firmware – Electronics, Lab measurement, Hardware bring-up, Sensing, Power subsystems, Motion subsystems, Embedded firmware, Real-time development, Control loops, Drivers, Timing, Interfaces

Software Development – Java, Spring Boot, Python, REST APIs, Docker, Linux, Git, Jenkins

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch)

Data Analysis & Visualization – Power BI, DAX, Tableau, R, pandas, NumPy, scikit-learn

Methodologies & Tools – Agile/Scrum, Jira, Confluence, Change-control management, UAT coordination

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin 02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India 01/2021

PROJECTS

Autonomous Drone Navigation System

- Developed core software modules for an autonomous drone navigation system, integrating **sensor fusion** algorithms (IMU, GPS, LiDAR) to achieve accurate real-time positioning.

- Implemented **path planning and obstacle avoidance** logic using Python and ROS, enabling the drone to navigate complex environments safely and efficiently.
- Designed and deployed a **data logging and telemetry** system to capture flight parameters and sensor data for post-mission analysis and performance tuning.

Industrial Automation Control Dashboard

- Built an interactive **Power BI dashboard** to monitor and control industrial automation equipment, visualizing real-time operational status, error logs, and key performance indicators.
- Developed **DAX measures** to calculate critical metrics such as uptime, throughput, and energy consumption, providing actionable insights for operations teams.
- Integrated with backend systems via **REST APIs** to ingest data from PLCs and manufacturing execution systems, ensuring a comprehensive view of the production floor.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)